MICRO-EAZY ECOSYSTEM · BLUEPRINT V2 · RE-COMPASSED AFTER LEGAL CLEARANCE

The Interchange

Consent, CRB 2.0, the closed learning loop, and the model no competitor can rebuild — because the advantage is the data, not the weights.

PREPARED BY	DATE	STATUS	SUPERSEDES
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01 What changed since v1

Counsel has cleared the plan. That removes the one blocking risk in v1 §8 and replaces it with a condition: consent must be provable, for every borrower, at every hop.

TABLE 1 · DECISIONS NOW LOCKED

QUESTION	DECISION	CONSEQUENCE FOR THE BUILD
Regulatory position	**Cleared, with conditions**	Consent Center becomes a Phase-1 component, not a Phase-4 one
Operating entity	**BirgenAI subsidiary**	Neutral branding matters more, not less — see §9
Data residency	**No constraint**	Cloudflare R2 confirmed; MinIO fallback dropped
Founding cohort	**Micromart · Axe · BuySimu**, plus qualifying entities on the shared `.198` book	Cohort assessment query pending approval
Commercial model	**Per-query above a contribution-linked free tier**	Metering and quota land in the Registry from day one
Infrastructure ownership	**You own all of it**	Nodes are hosted by you on your hardware; members get an endpoint, not a server
Build base	**LMS stack** — Next 15, Tailwind 4, Prisma	Vault cinematic ports forward; one stack across seven systems
The AI	**Retrieval + tools, plus own scoring models**	Ships in weeks; the moat is the pool, not the weights
Consent enforcement	**Hard gate — no `consent_ref`, no answer**	The network physically cannot serve an unconsented query
First ship	**Vault shell · Directory · Consent Center**	Demoable URL in ~2 weeks

ONE DECISION DESERVES A SECOND LOOK

You own all the infrastructure, and that is right for control, cost and speed. But note what it costs you rhetorically: v1 §4 argued that a lender joins because *their data never leaves their perimeter*. If you host every node, that sentence is no longer true, and a sophisticated competitor will notice. The tokenisation boundary in §4 is what preserves the argument — identity is destroyed before it reaches storage you control, so what you hold is unlinkable even by you. Say it that way, and the claim survives. Say "your data never leaves you" and it does not.

The Consent Center

Counsel's condition is the architecture's spine. Every borrower entering the ecosystem — through the Micro-Eazy PWA or through a member's own onboarding — agrees once, comprehensively, to everything the Interchange will ever do with their data. No agreement, no loan.

2.1 What the consent must cover

Consent is granted as **scopes**, each independently recorded and independently revocable. A loan requires the mandatory set; the optional set widens what the ecosystem can learn.

TABLE 2 · CONSENT SCOPES

SCOPE	WHAT IT AUTHORISES	REQUIRED FOR A LOAN
`mpesa.crunch`	Parse and derive features from the uploaded M-Pesa statement	**Mandatory**
`kyc.verify`	Identity verification, liveness, document capture, IPRS where available	**Mandatory**
`bureau.pull`	Query external CRBs (Metropol) on their behalf	**Mandatory**
`ecosystem.exposure`	Ask other members whether this person has live exposure, and answer such questions about them	**Mandatory**
`outcome.label`	Record how the loan ended, and use that label to improve scoring	**Mandatory**
`model.train`	Include their de-identified features in training data for ecosystem models and agents	**Mandatory**
`collections.contact`	Contact them, and share contactability signals with the member servicing the debt	Mandatory
`identity.disclose`	Reveal the lender's identity to a counterparty rather than aggregates only	Optional
`marketing.offers`	Receive pre-qualified offers from other ecosystem members	Optional

> **WHY BUNDLING THE MANDATORY SET IS DEFENSIBLE**
>
> Conditioning a loan on consent is lawful where the processing is genuinely necessary to provide the product — assessing whether to lend. The mandatory scopes above all serve that assessment, which is why they can be bundled. `model.train` is the one worth a second conversation with counsel: it is arguably necessary (the score that approves the next borrower is built from it) but it is also the one a regulator would probe hardest. Keeping the two marketing-adjacent scopes strictly optional is what demonstrates that the bundle is about lending, not about collecting everything you can.

2.2 The lifecycle

```
CAPTURE     PWA or member onboarding presents scopes, records affirmative act
            → device, IP, timestamp, wording version, signature/OTP evidence

ISSUE       POST /consent  →  { consent_ref, subject_token, scopes[], expires_at }
            consent_ref is an opaque handle; the ledger row is the evidence

VALIDATE    every Interchange call carries consent_ref
            Registry checks: exists · not expired · not revoked · scope covers
            this operation  → else 403 CONSENT_SCOPE_MISSING, logged both sides

REVOKE      borrower withdraws via PWA or by request to any member
            → propagates ecosystem-wide within one config cycle (≤15 min)
            → prospective only; existing labels retained under legitimate interest

PROVE       any member, any borrower, any regulator can be shown the exact
            wording agreed to, when, and every query it subsequently authorised
```

2.3 Two onboarding paths

Native — a borrower using the Micro-Eazy PWA uploads their M-Pesa statement as a mandatory step, then meets the consent screen before the application form. The `consent_ref` is minted inline and travels with every downstream call.

Bridged — a member onboarding on their own system must present equivalent wording and `POST /consent` to the Consent Center before their first query about that borrower. The Interchange supplies the canonical wording, a drop-in web component, and a REST endpoint. This is the integration burden of joining, and it is deliberately the *first* thing a new member builds — it establishes from day one that the network's currency is provable consent.

CRB 2.0

Metropol sees what regulated lenders chose to report, in monthly batches, about accounts that already exist. CRB 2.0 sees the ecosystem live, plus everything a bureau structurally cannot hold.

3.1 Where the advantage actually comes from

TABLE 3 · STRUCTURAL COMPARISON

DIMENSION	METROPOL / TRADITIONAL CRB	CRB 2.0
Freshness	Monthly submission cycle; weeks of lag	**Live** — queries hit the member's book
Coverage	Only regulated institutions that report	Every ecosystem member, reporting continuously
Cashflow	None — bureaus never see the bank rail	**M-Pesa statement features**, mandatory at application
Thin-file borrowers	“No record found” — the majority of first-time borrowers	Scored from cashflow behaviour alone
Rejected applicants	Invisible	**Observable** via ecosystem outcomes — see §5.2
Collections signal	None	1.34 M dispositions, 150 K PTPs, contactability
Intent	None	Application velocity across members — who is shopping right now
Cost per query	Per-report tariff	Free above a contribution-linked tier

> **WHAT CRB 2.0 IS NOT**
>
> It is not a replacement for Metropol and should never be sold as one. A regulated lender still has statutory reporting obligations to a licensed bureau, and Metropol sees banks and SACCOs that will never join this ecosystem. CRB 2.0 is a *complement* with a different shape: shallower on formal credit, far deeper on cashflow, intent and behaviour. The LMS default becomes CRB 2.0 because it is instant and free; Metropol remains available as an escalation for larger tickets.

3.2 Metropol-compatible by construction

The endpoint shape deliberately mirrors Metropol's — same request envelope, same `api_code` semantics, same report-type integers where an equivalent exists. A member already integrated with Metropol changes a base URL and a key pair, and keeps their parsing code. That is the cheapest possible migration path, and it is worth the constraint it imposes on our response design.

TABLE 4 · REPORT CATALOGUE

TYPE	REPORT	CONTENTS	METROPOL PARALLEL
1	Identity Verification	KYC status, liveness result, name/ID match confidence	Report 1
2	Delinquency Status	Worst bucket across the ecosystem, days past due	Report 2
3	Interchange Score	0–1000 network score with reason codes	Report 3 (Metro Score)

TYPE	REPORT	CONTENTS	METROPOL PARALLEL
11	Cashflow & Affordability	M-Pesa derived income, volatility, obligations, disposable estimate	Report 11, but from the rail
12	Full Enhanced	Everything below, assembled	Report 12
20	**Ecosystem Exposure**	Live multi-lender exposure, stacking velocity, 14-day new-credit count	*none*
21	**Intent Signal**	Applications across members in the last 30 days, approved and declined	*none*
22	**Contactability**	Best contact hour, PTP-keep rate, alternate numbers seen elsewhere	*none*
23	**Cohort Benchmark**	This borrower against their cohort across the ecosystem	*none*

3.3 Dynamic assembly and the PDF artefact

A report is not a fixed document — it is assembled from whichever **evidence blocks** exist for that borrower and whichever the requesting member has subscribed to. A borrower who has been KYC-verified, M-Pesa crunched, CRB pulled and scored yields a full report; a fresh applicant yields the blocks that exist, with the rest marked `unavailable` rather than fabricated.

```
EVIDENCE BLOCKS                    ASSEMBLY
————————————————                    ————————
identity.kyc           ┐            subscription tier
mpesa.cashflow         │            + consent scopes
bureau.metropol        ├──────────► + blocks present
scoring.pipelines[6]   │            ————————————————
repayment.behaviour    │            = this member's report
ecosystem.exposure     │                 │
collections.signal     ┘                 ├─► JSON  (API)
                                         └─► PDF   (console + storage)
```

Every generated report is written to Supabase Storage under `reports/{orgId}/{subject_token}/{report_type}/{iso}.pdf`, with the JSON payload retained beside it. Because `subject_token` is the key, a report stored by Micromart is retrievable by Axe under the same token when consent and subscription permit — which is what makes “request a customer's report from another member's LMS” work without either party exchanging a name.

04

The feature taxonomy

Everything the ecosystem knows about a borrower, organised into the families a model consumes. Roughly 120 features at launch, all computed on the member's side of the tokenisation boundary and published against `subject_token`.

TABLE 5 · FEATURE FAMILIES

FAMILY	COUNT	REPRESENTATIVE FEATURES	SOURCE
Identity & KYC	12	ID-name match confidence, liveness score, document quality, age, tenure since first seen, duplicate-device flag	PWA / LMS onboarding
M-Pesa cashflow	38	Median monthly inflow, inflow volatility, salary regularity, till/paybill ratio, night-transaction share, balance troughs, overdraft (Fuliza) reliance, betting share, airtime cadence	Statement cruncher
Application	10	Requested amount vs income, tenor, product, channel, time-of-day, form-completion time, edit count	Origination funnel
Bureau	14	Metropol score, account count by status, sector exposure, enquiry count, worst historical bucket	Metropol (when open)
Ecosystem exposure	16	Active loans, distinct lenders, outstanding band, stacking velocity 7/14/30d, time since newest disbursement, cross-member arrears	**Interchange only**
Repayment behaviour	18	On-time ratio, mean days late, worst bucket, early-settlement count, rollover count, instalment-to-inflow ratio, recovery-after-arrears rate	Serviceconnect
Collections	12	Contact attempts to resolution, PTP made vs kept, best contact hour, disposition mix, RO changes	CollectBox
Ecosystem intent	8	Applications across members 30d, decline count, approval-to-application ratio, shopping breadth	**Interchange only**

POINT-IN-TIME CORRECTNESS IS NOT OPTIONAL

Every feature must be computable *as it stood at decision time*. A model trained on “worst bucket ever” will look extraordinary in backtest and collapse in production, because at scoring time that value does not yet exist. This is the most common way a lending model dies, and the reason Plane B is built on Iceberg: its snapshot and time-travel semantics let the feature store reconstruct any borrower's vector as of any timestamp, cheaply and provably.

The closed learning loop

5.1 Why it has never actually closed

The nightly job in `emmanuel-arch/BirgenAI` that was supposed to label outcomes has failed every run since it was added. It is not a code defect — the workflow guards on two repository secrets, `APP_BASE_URL` and `CRON_SECRET`, and neither was ever set, so it exits 1 before making a request. **No loan outcome has ever been written back.** Every scoring record produced so far is unlabelled.

Setting the two secrets fixes the run. It does not fix the design. A GitHub Action firing an HTTP call at a Vercel app is a fragile place for the mechanism that decides what your models learn: it has no lineage, no retry semantics, no backfill window, no idempotency, and it fails silently in a repository nobody watches. In the target architecture, labelling is a **Dagster asset** owned by the Interchange, with the loan book as its upstream dependency and the label store as its output.

5.2 The reject-inference advantage

This is the strongest data-science argument for the entire ecosystem, and it should be the centrepiece of every member conversation.

A lender can only observe outcomes for loans it approved. Everyone it declined is an unlabelled hole in the training set. Train on approvals alone and the model learns to reproduce the existing credit policy rather than to predict repayment — the classic selection bias, and the reason in-house scorecards plateau. The standard remedies (reject inference by parcelling, augmentation, bureau-based performance proxies) are all attempts to *guess* what would have happened.

The Interchange does not have to guess. A borrower Micromart declined who was approved by Axe generates a real, observed outcome against the same `subject_token`. The rejected population becomes partially labelled with ground truth rather than inference.

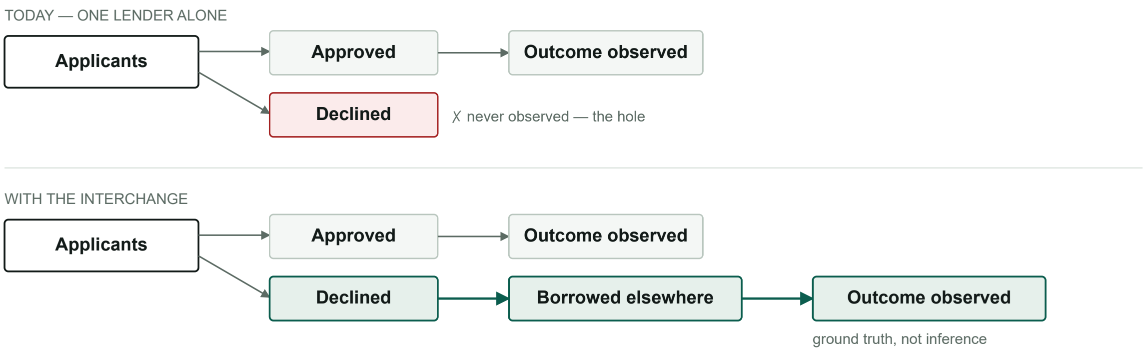

Figure 1 · The declined population is unlabelled for a lender operating alone, and partially labelled with observed ground truth inside the ecosystem. No bureau can sell this, and no single lender can build it.

5.3 The loop, as it should be built

1. **Decision snapshot.** At every credit decision, persist the exact feature vector, model version, score and outcome. This is the training row; it must be written synchronously, not reconstructed later.

2. **Outcome labelling.** A Dagster asset joins decisions to the loan book nightly and applies the label policy — `REPAID`, `DEFAULTED` at 90+ DPD, `IMMATURE` until the performance window closes. Idempotent, backfillable, with lineage.
3. **Ecosystem labelling.** Declined applications are matched by `subject_token` against outcomes reported by other members, producing the reject-inference rows from §5.2.
4. **Feature store.** Point-in-time vectors materialised into Iceberg gold tables, serving both training and inference from the same definition — the only reliable defence against training/serving skew.
5. **Training.** Gradient-boosted models (LightGBM) on the ~120 features, with monotonic constraints on features where direction is known, and reason codes derived from SHAP for regulatory explainability.
6. **Champion / challenger.** New models shadow-score live traffic without deciding, until they beat the champion on a held-out window. Promotion is a config change, and reversible.
7. **Drift monitoring.** Population stability on every feature family, alerting when an input distribution moves — which is how you find out that a member changed their onboarding form before it silently corrupts the score.

> **DO THIS BEFORE ANYTHING ELSE IN THE LOOP**
>
> Set `APP_BASE_URL` and `CRON_SECRET` in the BirgenAI repository today, so labelling starts accumulating while the Dagster version is built. Every night it stays broken is a night of decisions that can never be labelled — the feature vectors were never snapshotted, so that data is not delayed, it is *gone*.

ServiceSuite AI & the Interchange model

The advantage is the data and the tools, not the weights. Said plainly: a competitor with the same model gets nothing, and a competitor with the same data gets everything.

The instinct to "train a model no Kenyan can replicate" is right about the goal and worth redirecting on the method. Fine-tuning a language model on ecosystem data produces something proprietary but weaker at reasoning than a frontier model, needs GPU spend and retraining discipline, and — critically — *the fine-tune is not the moat*. The moat is that only you can answer "how many other lenders is this person servicing right now", because only you have the network. Put a frontier model on top of tools that reach that data and no competitor can match the answers, whatever model they run.

6.1 Two layers

TABLE 6 · THE AI STACK

LAYER	TECHNOLOGY	JOB
Conversational	Claude + tool calling	"Ask ServiceSuite AI" in every system. Answers questions about a borrower, a branch, a cohort or the ecosystem by calling Interchange tools — never by recalling training data.
Predictive	LightGBM ensembles	Score, probability of default, PTP propensity, best contact hour. Trained on the ~120 features, retrained by the closed loop, explained with SHAP reason codes.

6.2 The tools the model is given

```
interchange.exposure(subject_token)        live multi-lender exposure
interchange.report(subject_token, type)    any CRB 2.0 report
interchange.features(subject_token)        the ~120-feature vector
interchange.score(subject_token)           score + SHAP reason codes
interchange.cohort(filter)                 anonymised peer comparison
interchange.benchmark(org_id, metric)      this member vs ecosystem median
lms.customer360(customer_id)               the member's own book
collectbox.history(loan_id)                dispositions, PTPs, contact log

EVERY tool call carries the caller's org_id and the borrower's consent_ref.
The tool layer — not the prompt — enforces scope. A model cannot be
talked into returning data the consent does not cover.
```

GUARDRAIL THAT MUST NOT BE SKIPPED

Consent and access control belong in the tool implementations, never in the system prompt. A prompt instruction is a suggestion to a language model; a check in the tool is a wall. This distinction is what makes "Ask ServiceSuite AI" safe to put in front of a competitor's staff, and it is the first thing a security reviewer will test.

Deployment mirrors the existing pattern: the assistant already lives bottom-right in the LMS console. The same component ships in all seven systems, with the `interchange` model selectable wherever the caller's consent and subscription permit. From Customer 360 a loan officer summons it and asks questions no other assistant in Kenya can answer — because no other assistant is wired to this pool.

07 Metropol — current status

BLOCKED ON METROPOL, NOT ON US

The production key pair is genuine and the integration is complete. Two faults were stacked: our config defaulted to the *test* port 5555 because `METROPOL_PORT` is absent from `.env`, which produced E003 on every service; and the correct production port **22225 is firewalled** from our network. Probe of 21 Aug 2026 — `5555` reachable, `22225` and `443` no route. That is source-IP allow-listing, the one question Metropol has not answered.

Action: ask Ambale to allow-list `41.90.172.22` now, and the Vercel production egress ranges before go-live. Set `METROPOL_PORT=22225` in `.env` in the same change.

Note for whoever runs the first live pull: Report 1 keys on **national ID**. The mobile-keyed report is **Report 11** (`/report/creditinfo/mobile`). An MSISDN sent to Report 1 will fail regardless of entitlement.

This does not block the build. CRB 2.0's `bureau.metropol` evidence block is simply marked `unavailable` until the port opens, and every other block — KYC, cashflow, scoring, repayment, exposure — is computed from data you already hold.

08

Build plan

TABLE 7 · SPRINTS

SPRINT	WINDOW	DELIVERABLE	DONE WHEN
0	Now	Set the two BirgenAI repo secrets. Set `METROPOL_PORT`. Email Ambale re allow-listing. Run the `.198` cohort assessment.	Labelling accumulates; cohort shortlisted
1	Weeks 1–2	Interchange scaffold on the LMS stack. Vault shell ported. Member Directory. **Consent Center** with ledger, scopes, revocation.	`interchange.servicesuitecloud.com` live and demoable
2	Weeks 3–5	Registry: member registry, OPRF token service, policy engine, metering. Message log with hash chain.	Two members exchange a signed, consented, logged call
3	Weeks 6–8	Exposure engine — fan-out, Bloom filters, aggregation. CRB 2.0 report types 1, 2, 20.	p95 under 400 ms on the real Micromart book
4	Weeks 9–12	Plane B: CDC from Micromart, Iceberg on R2, feature store, decision snapshots, Dagster labelling incl. reject inference.	A model trains on point-in-time vectors
5	Months 4–5	Interchange Score, full CRB 2.0 catalogue, PDF artefacts, ServiceSuite AI with the `interchange` model.	A loan officer asks a question only this network can answer
6	Months 6+	Members beyond the founding cohort. Governance entity. Neutral domain.	Ten contributing members

8.1 Repository layout

```
github.com/emmanuel-arch/Interchange

  apps/
    console/         Next 15 · the member console (vault shell, directory,
                     consent, audit, CRB 2.0 reports)
    registry/        member registry · OPRF · policy · metering · message log
  packages/
    envelope/        X-Road-compatible message envelope + signing
    consent/         consent SDK: web component + REST client for members
    features/        the ~120-feature definitions, shared by training & serving
    ui/              ported vault cinematic, shared with the six systems
  data/
    dagster/         labelling, reject inference, feature materialisation
    dbt/             bronze → silver → gold on Iceberg
  docs/              this blueprint, the data contracts, the consent wording
```

From the GoldStrike import: `my-app/app/auth/login`, `components/screensaver` and `components/network` port forward into `packages/ui`. The Python trading engine, and every trading route — trades, backtest, engine, investors, telegram, payments — are deleted rather than adapted; nothing in them maps onto this domain.

09

What still needs a decision

1. **The `model.train` scope.** Put the exact wording to counsel. It is the scope most likely to be challenged, and the one the entire ML programme depends on.

2. **Cohort qualification threshold.** Pending the `.198` assessment — my recommendation is a minimum live book and a recent disbursement, not raw history, since a dormant book contributes nothing to exposure.
3. **Free-tier boundary.** How many queries does contribution buy? Suggest indexing it to contribution volume so it self-balances rather than needing renegotiation per member.
4. **Neutral domain timing.** Owning the infrastructure makes the hosting neutral in fact but not in appearance. Decide when `interchange.africa` becomes primary.

> **IMMEDIATE NEXT STEP**
>
> Sprint 0 is four small tasks and none of them need a decision. The two repository secrets in particular should be set today — every night they stay unset is a night of credit decisions whose feature vectors are never captured, and that loss is permanent.